The Flower: Visualizing the Solved-Move Landscape of the Rank-48 4×4 Scheme, and the Search Heuristics It Yields

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Research note — logic repo (https://github.com/gsidebottom/logic), matmul track, 2026-07-09. Companion to "Rigidity of the Rank-48 4×4 Matrix-Multiplication Scheme under Solved Flip Moves" (`doc/matmul_rigid48_paper.md`), which proves the theorem this note visualizes, and to the 3×3 additive-complexity paper (`doc/matmul_adds_paper.md`). Interactive version of Figure 1: `https://claude.ai/code/artifact/e2167bb4-fb0d-4295-a0f9-b4554142e39d`. Every number below has a mechanical check (§6).

Abstract

The companion paper proves that the rational rank-48 scheme for 4×4 matrix multiplication — the only rank-48 scheme usable over the real numbers — is *rigid*: the graph of states reachable by one split and solved flips is finite (7,408 states) and contains no reduction except the trivial return to the seed. This note treats that certified graph as **data**. We export it with per-state search metrics, visualize it, and read off its anatomy, which turns out to be strikingly simple: a **flower** — a single hub (the seed), 6,488 pendant leaves that admit no move except undoing themselves, and a thin 920-state *fringe* where all nontrivial structure lives and where every local search signal saturates almost immediately.

From the picture we derive six concrete search heuristics, and they pay off at once: a gradient signal (**nearmiss**, the number of solvable one-move coincidence targets) that is provably capped at 2 inside the certified component **deepens to 7 and then 14** one split level further out and, under seven levels of beam guidance, reaches a mean of 192 — 48× a matched-rank random control, certifying the gradient as heritable structure rather than rank inflation; a 19× budget-focusing rule; and a cost-recalibration discipline that caught a uniform two-split enumeration mispriced by two orders of magnitude (killed at 3% after 18 CPU-hours; its true cost ≈500 CPU-hours) and replaced it with a guided program that covers the region all metrics favor at a fraction of the price. Whether the deepening gradient ever converts into an actual rank reduction — a new rank-48 scheme, or the characteristic-0 rank-47 that would be a record — is the open question the running campaigns address.

1 The graph

All terms from the companion paper; briefly: the **seed** is the Dumas–Pernet–Sedoglavic rational $\langle 4 \times 4 \times 4 : 48 \rangle$ decomposition (48 rank-one summands $a \otimes b \otimes c$ over $\mathbb{Z}[\frac{1}{2}]$); a **split** replaces a summand by two (rank $+1$, and over $\mathbb{Q}$ can target *any* partner's factor); a **solved flip** is a flip whose scalar λ is chosen by solving $f_i + \lambda f_j = \mu f_m$ — the only way exact coincidences occur over the rationals; a **reduction** merges two summands proportional in two slots (rank -1).

`flip48 --graph` exports the full 1-split component: **7,409 nodes** (the seed plus 7,408 rank-49 states — independently matching the count pinned in the Lean theorem) and **16,000 edges** (6,768 splits, 2,328 solved flips, 6,904 reductions), with four metrics per state:

- **shared** — factor pairs equal in some slot (flip eligibility);
- **coinc** — solved flips available;

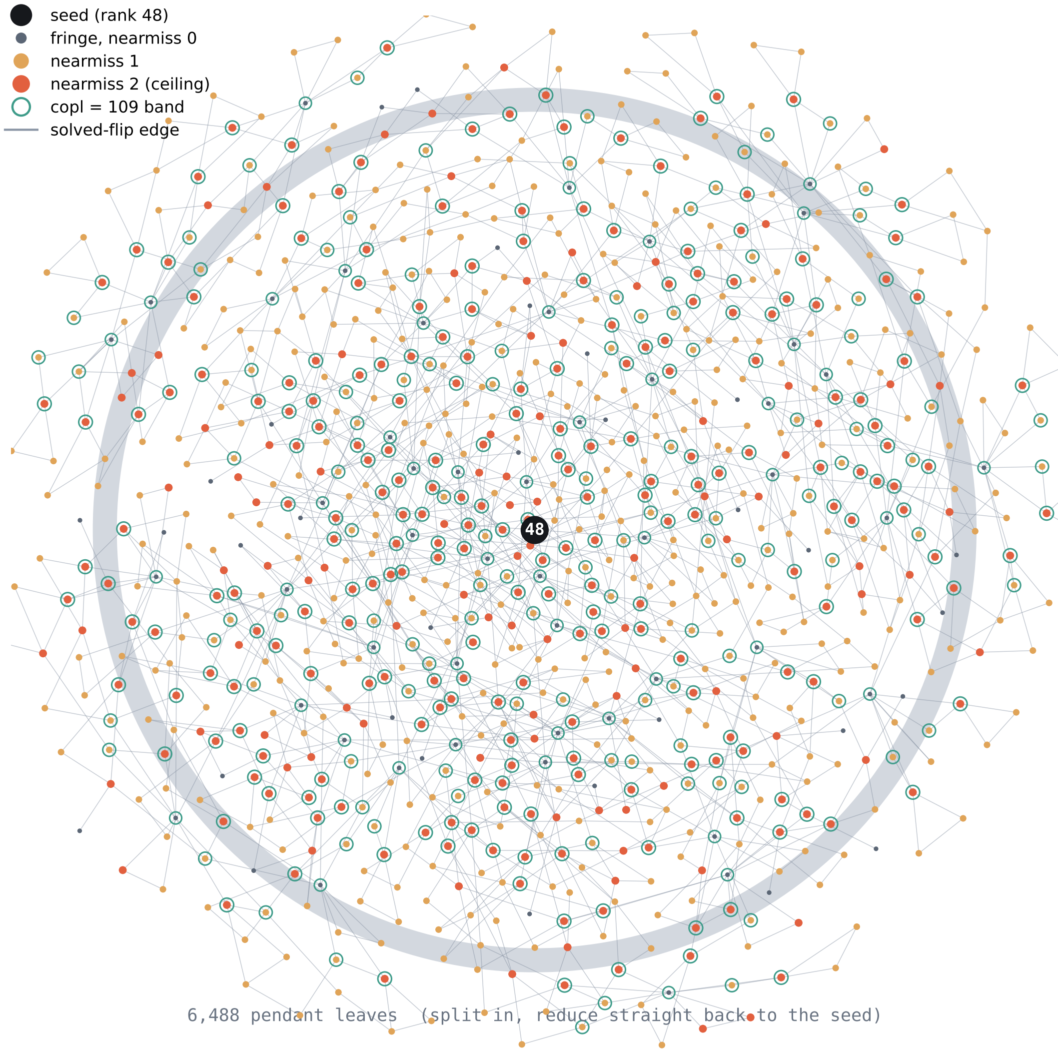

Figure 1: The solved-move component. Hub = seed; annulus = 6,488 pendant leaves (aggregated); scattered structure = the 920-state flip-active fringe, colored by nearmiss, teal-ringed where copl = 109.

- **nearmiss** — distinct third summands reachable by a ply-1 coincidence: how many partners this state could become aligned with in one move. A reduction needs alignment with the *same* partner in *two* slots, so nearmiss counts the doors that a second aligned slot would turn into exits;
- **copl** — coplanar factor triples in the a-slot (rank-2 triples): the raw material from which solvable moves are made.

2 Anatomy: the flower

measurement	value
states / edges	7,409 / 16,000
reductions → seed	**6,904 of 6,904**
degree-2 pendant leaves	6,488 (88%)
flip-active fringe	920 states, 1,468 flip edges
flip-clusters of the fringe	**16**, sizes 53–61 ($2{\cdot}53 + 4{\cdot}55 + 2{\cdot}57 + 4{\cdot}59 + 4{\cdot}61 = 920$)
nearmiss distribution	0: 6,568 · 1: 516 · 2: 324 — **max 2**
coinc distribution	0: 6,568 · 2: 516 · 4: 324
copl distribution	63: 2,128 · 64: 4,896 · **109: 384**
max coefficient anywhere	6 (magnitudes never grow)

Four readings:

1. **The rigidity theorem, as one number.** Every one of the 6,904 reduction edges points at the seed. The certified statement "no reduction except the trivial return" is *visible.*

2. **The component is almost entirely inert.** 88% of states are petals — one move in, one move out, no interaction with anything. Uniform search effort is 88% wasted by construction.

3. **The fringe is where structure lives, and its signals saturate.** Only 920 states have any solved flip at all; no state has more than 2 nearmiss targets or 4 available moves; and one distinguished band — 384 states with 109 coplanar triples against the 63–64 baseline — concentrates the geometric richness. Inside the certified component there is, quite literally, no gradient to climb: the landscape is flat at height 2.

4. **The fringe has sixteen-fold substructure.** Its 920 states decompose into exactly 16 flip-connected clusters of near-equal size (53–61 states). Equivalently, the seed is a cut vertex of spectacular degree: every split and reduction edge is incident to it, so deleting it isolates the 6,488 leaves completely and shatters all remaining connectivity into those 16 near-equal components. Near-uniformity like that is unlikely to be accidental: the scheme's automorphisms act on the component, and the clusters look like an orbit decomposition — observed, not yet proven. Aggregating each cluster's *directed* flip graph by strongly connected components sharpens this: same-size clusters have **identical condensation profiles** (SCC size/degree/nearmiss multisets and typed edge signatures; five species with 15/16/17/22/19 SCCs), every condensation is exactly three layers deep, SCC sizes are only 1, 2, 4, 6, 8, and there are **no trap cycles** — the only dead ends are the 80 singleton sinks (the coinc-0 states; 664 of the 1,468 flip pairs are one-way). A companion figure, the *condensation atlas* (`doc/fig_flip_atlas.html`), draws the five species and the chase-vs-null trajectory. Flips are also where the metrics move: 74% of flip edges connect states of *different* nearmiss (308 of them jump $0 \leftrightarrow 2$ in a single move), and 78% touch the copl-109 band. In the interactive figure, hovering any state or flip edge isolates its cluster and displays the per-edge gradient.

3 The heuristics

H1 — Spend search where the structure is (19× focusing). Second splits from pendant leaves recreate the situation the certificate already closed. Restricting second splits to the 920-state fringe (and ordering by nearmiss, then the copl-109 band) concentrates the budget on ∼12% of parents with, empirically, all of the signal.

H2 — Never walk randomly; only solved moves. (Restated from the companion paper as an operating rule.) Ten million random-λ moves produced zero coincidences; exact events over $\mathbb{Q}$ are measure-zero. Every productive move is the solution of a small linear system — search over $\mathbb{Q}$ is *constructive*, not stochastic.

H3 — nearmiss is the gradient. It measures exactly the precondition of the goal event (a two-slot alignment with a single partner). Its ceiling inside the certified component is 2; the question "is the landscape flat everywhere, or only here?" is decided by measuring it at depth. Measured: **depth-1 max 2 → sampled depth-2 max 7 → guided beam level-1 max 14 (mean 11.1 over a 1,500-state beam)**. The gradient is real, and it steepens under guidance — the strongest empirical argument yet that the rigidity boundary is a property of the *radius*, not of the whole landscape.

H3, tested against the null. A climbing nearmiss curve has a cheap alternative explanation: a depth-k state has $48+k$ summands, so the count of possible alignments could inflate mechanically with rank, and a beam would then be "climbing" a staircase that rises under everyone's feet. The control (`--nmrand`): at each depth, 5,000 *unguided* random split-walks from the seed, nearmiss of each raw endpoint — same split distribution as the beam, no selection, no inheritance. Matched at equal rank:

rank	beam mean	beam max	rnd mean	rnd p99	rnd max (5,000)	top-1,500 of 5,000
50	5.1	9	0.21	5	9	0.7
51	14.3	29	0.51	6	12	1.7
52	27.9	50	0.92	7	17	3.0
53	51.1	76	1.55	10	27	4.6
54	85.3	122	2.22	12	20	6.0
55	137.5	194	3.02	14	25	7.3
56	**192.3**	282	4.01	17	**33**	9.0

Rank inflation exists but is tiny (unguided mean reaches only 10.5 by rank 61, the campaign's terminal depth — the beam passed that at level 2). At rank 56 the beam's *mean* is 48× the unguided mean, 11× the unguided 99th percentile, and nearly 6× the unguided *maximum over 5,000 draws*; one-shot selection without guidance (best 1,500 of 5,000) achieves 9.0, i.e. 5% of the beam mean. A typical beam state outscores the best unguided state of equal rank — no one-shot selection from the unguided distribution produces that. The only mechanism left is inheritance: high-nearmiss parents beget higher-nearmiss children. And the beam/null ratio itself climbs (24× → 28× → 30× → 33× → 38× → 46× → 48× across levels 1–7): guidance compounds. **The gradient is heritable structure, not a rank artifact.**

H4 — The copl band is a prior. Coplanarity is the raw material of solvable moves; the 384-state copl-109 band is the natural first target for any deeper enumeration, and band membership is computable before committing any search budget to a parent.

H5 — Measure closure inflation before exhausting. The 1-split closures average 2.3 states; a uniform 2-split enumeration priced at that figure looked like a ∼10-CPU-hour job. Direct measurement showed level-2 closures average ∼580 **states (250×)** — repricing the uniform sweep at ∼500 CPU-hours. It was killed at 3% (18 CPU-hours spent) and replaced with the guided program below. The rule: before exhausting level $k+1$, measure its branching on a sample; the extrapolation from level k is not a bound, it is a guess.

H6 — A certificate needs an audit trail, not a counter. The first fringe-exhaustive campaign hit its state budget at "∼644 of 840 parents done" — and could not say *which* 644. Its parents were processed by a work-stealing parallel loop, so the completed set was a scattered, nondeterministic subset, not a prefix (a 2-parent test run completed indices 76 and 2,775); a planned prefix-skip resume would have silently left coverage holes in the certificate, and was retracted before use. The rule: any enumeration whose value is the claim "*all* of X was covered" must log the *identities* of completed work (stable across runs), not a count — completeness of coverage is otherwise unrecoverable from the survivor. The engine now appends each finished parent index to `fringe_done.txt`, and `--resume` skips exactly the logged set; campaign 1's 4.02 billion states stand as evidence, campaign 2 is the certificate run.

4 The running program, and the open question

Two campaigns implement the heuristics (both with progress tickers, global state budgets, and inline exact verification of any find):

- **Fringe-exhaustive** (`--pursue5 0 --fringe-only`): *all* 7,056 second splits of each of the 840 gradient-positive parents, every root closed under solved flips with reduction continuations. **Complete: all 840 parents exhausted — zero reductions found, depth-2 nearmiss ceiling 14.** The certificate took $\approx$9.4 billion states across three runs: campaign 1's scattered $\sim$644 parents at its 4×10^9 budget stop (unusable for coverage — see H6), then the completion-logged campaign 2 in two runs (306 + 534 parents, 3.44×10^9 states in the final run), bridged losslessly across a host interruption by the H6 machinery. The rigidity certificate now extends to: **no reduction is reachable through the fringe at 2-split radius.**
- **Gradient chase** (`--pursue6`): best-first beam on nearmiss across split depths (beam 1,500, 60 sampled splits per frontier state, closure cap 150). Through level 7 (rank 56): nearmiss max 282, beam mean 192.3, with the per-level mean rising (+9.2, +13.6, +23.2, +34.2, +52.2, +54.8) while the unguided baseline crawls (+0.3, +0.4, +0.6, +0.7, +0.8, +1.0) — and the H3 control certifies the climb is not rank inflation. (Operational note: a silent default state budget ended the first run after level 6; it was resumed from the level-6 frontier dump, which under the then-current format kept only the top 50 of the 1,500-state beam. Dumps now persist the full frontier, so future stops resume losslessly. The near-flat acceleration at the reseed level — +52.2 $\to$ +54.8 — may be an artifact of that narrowed beam; level 8, grown from the full beam again, decides.) Yet through 0.82 billion states, **zero conversions**: hundreds of one-slot alignments available per frontier state and not one same-partner second-slot event has ever fired. The chase asks the one question that matters: **does nearmiss keep climbing until a double-coincidence actually fires, or is there an obstruction — a radius-independent invariant separating alignment from reduction?** A conversion produces a genuinely new rank-48 scheme (fresh material for the operation-count program of the companion papers) or a characteristic-0 rank-47 — a record. A plateau — or a curve that climbs forever without converting — maps the next rigidity boundary.

5 Why visualize at all

The honest answer: every quantitative decision in §3 was *available* in raw logs, and none of it was *seen* until the graph was drawn. The flower shape — hub, petals, thin fringe — made three facts simultaneously obvious that tables had kept separate: reductions all point home; almost everything is inert; the interesting minority is small enough to exhaust. One picture converted a mispriced brute-force plan into a guided program and surfaced the gradient question that is now the campaign's spine. For search spaces born from exact algebra, where every state is expensive to think about individually, a faithful picture of the *whole* is the cheapest instrument there is.

6 Reproduction

```
cargo build --release --bin flip48
./target/release/flip48 --graph matmul/dps48/graph48.json --cap 256   # 17 s
# anatomy numbers: any JSON tool over graph48.json (see the table)
./target/release/flip48 --pursue5 0 --fringe-only --threads 8 \
    --budget 7000000000            # fringe-exhaustive campaign
# completed parents log to found48q/fringe_done.txt; add --resume
# after any interruption to skip exactly the logged set (H6)
./target/release/flip48 --pursue6 --beam 1500 --samples 60 --depth 8 \
    --threads 4                    # gradient chase
./target/release/flip48 --nmrand 13 --n 5000 --threads 4   # H3 null
./target/release/flip48 --worked                # Appendix A trace
```

The interactive Figure 1 (hover any fringe state or flip edge for its metrics and per-edge gradient; hovering isolates the state's flip-cluster) is published at the artifact URL in the front matter and archived in-repo as `doc/fig_flower_interactive.html` (self-contained — open it in any browser); the static figure is regenerable from `graph48.json` by the script in the repository history.

A A worked walk: seed → fringe → sink

One concrete walk from the seed through solved flips, visiting nearmiss 1, 2, 1, 0 and the copl-109 band — generated deterministically by `flip48 --worked` (the first split in scan order whose flip-cluster admits the full milestone visit), with every state re-verified exactly (128-bit integer arithmetic) against the 4×4 tensor. Notation: e_{rc} is the 4×4 matrix unit (row r, column c); a summand is $a \otimes b \otimes c$ in canonical gauge (a, b primitive integer vectors; the dyadic scalar rides on c).

The seed is flip-isolated (shared 0, coinc 0, nearmiss 0, copl 16): no two of its 48 summands share a factor in any slot, so no flip applies anywhere. The only move in the system is a split.

The split (rank 48 → 49). Summand 0 against summand 25 in the b-slot, with $b_0 = e_{31} + e_{41}$ and $b_{25} = e_{32} + e_{33} + e_{34} + e_{42} + e_{43} + e_{44}$:

$$a_0 \otimes b_0 \otimes c_0 \;=\; a_0 \otimes b_{25} \otimes c_0 \;+\; a_0 \otimes (b_0 - b_{25}) \otimes c_0.$$

The first part overwrites summand 0; the remainder becomes summand 48. This *engineers* alignment: pair $(0, 25)$ now agrees in slot b, and pair $(0, 48)$ agrees in slots a and c — the latter being exactly the split-undo reduction, the trivial return the rigidity theorem allows. Root metrics: shared 3, coinc 2, **nearmiss 1**, copl 64.

Step 1 — up to nearmiss 2, into the copl-109 band. The shared pair $(25, 0)$ in slot b opens coincidence systems in the other two slots: solve $a_{25} + \lambda\, a_0 \propto a_m$ over dyadic λ and all third summands m (a 2×2 Cramer system per candidate, then a 16-coordinate exact check). A solution exists at $m = 27$, $\lambda = -1$:

$$\underbrace{\begin{pmatrix} 1 & -1 & 1 & 1 \\ 1 & -1 & 1 & 1 \\ 1 & -1 & 1 & 1 \\ -1 & 1 & -1 & -1 \end{pmatrix}}_{a_{25}} - \underbrace{\begin{pmatrix} 1 & -1 & -1 & -1 \\ 1 & -1 & -1 & -1 \\ -1 & 1 & 1 & 1 \\ 1 & -1 & -1 & -1 \end{pmatrix}}_{a_0} = 2 \underbrace{\begin{pmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & -1 & 0 & 0 \\ -1 & 1 & 0 & 0 \end{pmatrix}}_{a_{27}}.$$

After the flip (gauge folds the factor 2 into the c-slot), summand 25's a-factor *equals* a_{27} — a new shared pair $(25, 27)$ in slot a. The compensating rewrite lands on c_0 (it gains $\pm\frac{1}{4} e_{33}/e_{43}$ terms), preserving the tensor sum exactly. New state: shared 3, coinc 4, **nearmiss 2, copl 109** — one flip doubled the door count and entered the band.

Step 2 — back down to nearmiss 1. Pair $(48, 0)$ shares slot a; solving in slot c finds $c_{48} + (-1)\, c_0 \propto c_{25} = \frac{1}{2} e_{33} - \frac{1}{2} e_{43}$, and after the flip $c_{48} = -\frac{1}{4} e_{33} + \frac{1}{4} e_{43}$. The compensation restores b_0 to its original $e_{31} + e_{41}$ — the walk is circling near its origin. State: shared 2, coinc 2, nearmiss 1, copl 109.

Step 3 — into a flip sink. Pair $(48, 0)$ shares slot a again; solving in slot b lands b_{48} exactly on b_{25}. Final state: **shared 3, coinc 0, nearmiss 0**, copl 109 — three aligned pairs and *not one solvable coincidence*. Alignment without exits: the concrete face of the alignment-vs-reduction gap of §4. A reduction would need a single partner aligned in *two* slots simultaneously; here every alignment lands on a different partner–slot combination (the sink's pairs are $(48, 25)_b$, $(48, 0)_a$, $(25, 27)_a$), and the walk ends.

state	move	shared	coinc	nearmiss	copl
seed (rank 48)	—	0	0	0	16
root (rank 49)	split $0 \leftarrow 25$ in slot b	3	2	1	64
s_1	flip $(25, 0)$: a-slot, $\lambda = -1$, target 27	3	4	**2**	**109**
s_2	flip $(48, 0)$: c-slot, $\lambda = -1$, target 25	2	2	1	109
s_3	flip $(48, 0)$: b-slot, $\lambda = -1$, target 25	3	**0**	**0**	109

Throughout, coefficient magnitudes never exceed 2 and denominators never exceed 4, matching the magnitude row of the anatomy table. Two incidental observations. First, all three flips involve only the split participants $\{0, 25, 48\}$ and a single outside target (27) — solved moves are local. Second, the seed's coplanar count (16) coincides with the number of flip-clusters (16); we have not pursued whether that is more than numerology.

Acknowledgments

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References

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